

Lia — Session Summary

The whole build, one line at a time.

Every exchange in the build of Lia, one line for what was asked and a short account of what came of it. Times and tool counts come from Claude Code's session log; the full wording is in *Lia — Session Transcript*.

58 exchanges	1025 tool calls	01 Aug, 12:52 started	~30 hrs elapsed
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Getting her talking

- | | |
|---------------------------------------|--|
| 1
01 Aug, 12:52
10 tools | Introduced the project; stuck because the editor's Run button can't take typed input.
Confirmed the code was fine by running it end to end. The fix was a real terminal. Flagged that DB_PATH was relative, so launching from the wrong folder would silently create a second, empty memory. |
| 2
01 Aug, 13:26
24 tools | Typed something and nothing came back.
Not broken — silent . Replies were fetched whole before printing, so ~20s passed with a blank screen. Added token streaming and progress markers. Also found fact extraction was storing nothing, and made DB_PATH absolute. |
| 3
01 Aug, 13:41
34 tools | Rename Aurora to LIA; add voice, with room to add my own voices later.
Built the whole voice layer: Piper for speech, a voices/ folder you drop .onnx files into, sentence-chunked speech so she starts talking before the reply finishes, and Whisper for listening. |
| 4
01 Aug, 13:58
10 tools | Fix the two things you flagged.
Diary was narrating as if it were the human — caused by user:/assistant: labels. Relabelled to Them/Lia . Added a deterministic filter so junk like name: "you" can't overwrite real facts. |
| 5
01 Aug, 14:03
5 tools | Analyse the project.
Found the biggest quality flaw: half her recall was her own past replies , unlabelled, which she read back as fact. Also: diary written but never read, no sense of time, embeddings stored as JSON (10× bloat). Measured the scan cost and said plainly it wasn't worth optimising yet. |
| 6
01 Aug, 14:06
2 tools | Fix line 8, column 16.
Type checker couldn't verify sys.stdout.reconfigure. Switched to getattr, which is also more honest about why the guard exists. |
| 7
01 Aug, 14:10
5 tools | Is everything fine? Explain the directories.
All modules compiled and ran. Caught her confabulating in that very run — inventing that you'd been nervous. Mapped the folder layout and the strict layering. |
| 8
01 Aug, 14:13
1 tools | How do I run it?
The command, what to expect, and why to type <i>bye</i> rather than closing the window. |

Giving her a voice

- 9**
01 Aug, 14:22
10 tools
- Set it up so I can talk to her by voice.**
Upgraded to `small.en` after testing showed `base.en` heard “*Ania*” for “*Anaya*”. Biased recognition using names she already knows, which fixed her own name coming out as “*Leah*”.
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- 10**
01 Aug, 14:31
2 tools
- What if she were an app I just open my laptop to?**
Measured the real gap: cold start 9.3s vs 2.4s warm. Laid out five things standing in the way and staged them, flagging that **sessions never ending** would mean an always-on Lia never learns.
-
- 11**
01 Aug, 14:39
26 tools
- Build stage 1.**
Models kept warm, sessions that close themselves after 12 quiet minutes, launcher and autostart script. Found the hidden bottleneck: Ollama's default 32k context was running **63% of her on CPU**. Pinning 8192 cut reply time by a quarter.
-
- 12**
01 Aug, 15:54
31 tools
- Do the retrieval fix, then stage 2.**
Retrieval limited to your words with timestamps. Added open-mic listening with Silero VAD, keyboard-vs-mic arbitration, and mic gating. Fixed her reading stage directions aloud.
-

Turning her into an app

- 13**
01 Aug, 16:03
14 tools
- Make her an app I can add to startup.**
Tray app with no window — only possible because voice detection removed the need for a keyboard. Verified quitting still writes her diary first.
-
- 14**
01 Aug, 16:15
49 tools
- Add a wake word, an exe, and tell me how to start.**
Wake word matched **on the transcript** rather than a trained model — no dataset needed, and it tolerates Whisper's spellings. Built the exe. Made paths survive being frozen.
-
- 15**
01 Aug, 19:57
1 tools
- How do I get her into startup?**
The one command, and what it actually does.
-
- 16**
01 Aug, 19:58
1 tools
- If I restart now, will she be there?**
No — I'd built the installer but deliberately not run it. Said so plainly.
-
- 17**
01 Aug, 20:00
5 tools
- Do it.**
Installed. Hit a real bug of my own: PowerShell variable names are case-insensitive, so my local `$exe` collided with the `-Exe` switch. Verified by launching the shortcut.
-
- 18**
01 Aug, 20:03
1 tools
- Confirm she'll be there after a restart.**
Yes — both she and Ollama were in the Startup folder.
-

Making her yours

19	Greet me, call me Warlock, and can she learn things like Morse code?
01 Aug, 20:05 17 tools	Greeting generated from her diary, which finally made the diary useful. Tested Morse honestly: she fails even with a perfect reference table in front of her . Not a memory problem — tokenisation. The fix would be code, not teaching.
20	A folder for PDFs, and make her voice softer.
01 Aug, 20:19 36 tools	Built the library with page-level citation. Stripped repeated page headers that were poisoning every search. Played four voice options rather than guessing.
21	Amy as she was, and make sure she says Warlock.
01 Aug, 20:54 10 tools	Reverted the softening. Found the name had reverted on its own — extraction re-derived the old one from stored turns. Added protected facts and purged the stale references.
22	Thanks.
01 Aug, 21:01 0 tools	—
23	Can I say “Lia stop” to stop her listening?
01 Aug, 21:04 1 tools	No. It would just become conversation. And she can't hear you at all while speaking. Said so rather than implying it half-worked.
24	What do you mean?
01 Aug, 21:08 0 tools	Re-explained without jargon: what barge-in is, why speakers break it, why headphones don't.
25	I use both.
01 Aug, 21:09 18 tools	Built spoken commands, and made interrupting work on speakers too — she compares what she hears against what she's currently saying , so her own voice is recognised and ignored.

Fixing what the real world broke

26	She didn't respond after a restart, and the diary won't open.
01 Aug, 21:34 33 tools	Three causes: she spent four minutes re-reading a PDF before speaking (absolute paths made it look new), speech recognition loaded after the greeting, and ignored speech was logged nowhere. The diary was fine — there was simply nothing to write yet.
27	I didn't want her going through PDFs on her own. And remove Anaya and Kamala.
01 Aug, 22:13 12 tools	Made the library strictly on-demand. Removed all of it — and said plainly that I had invented Anaya, Kamala, Maya and Biscuit during my own testing . Undid my overreach of having her read your PDF unasked.
28	The tray boxes were unchecked and she's not responding.
01 Aug, 22:33 18 tools	She'd asked how your day was, then ignored “hello” — my overcorrection from the previous fix. Gave the greeting a 30s reply window. The unchecked boxes were real: the Ollama wait blocked before the tray was told her state.
29	“Failed to import embedded python interpreter.”
02 Aug, 07:15 7 tools	You'd launched her mid-rebuild — my fault for rebuilding while she was in your startup. Builds now stage and swap, so a failed build leaves the working app untouched.

30

02 Aug, 07:24
1 tools

Should Ollama stay in startup?

Yes, and it's safer — when Windows starts it, it never touches her folder. Cost measured at ~130 MB RAM and no GPU while idle.

31

02 Aug, 07:26
14 tools

Is there a better approach than that trade-off?

Yes — my framing was wrong. The keep-alive timer **resets on every request**, so it only governs idle time. She now releases the models when a conversation ends and starts reloading the moment you speak.

Writing it down

32

02 Aug, 07:39
11 tools

Make a PDF of the changes, and a beginner's guide.

Both written, including the limitations rather than only the wins.

33

02 Aug, 09:23
1 tools

Is the project completed?

Working and deployed, but not finished: no way to forget, still embroiders, no tests, drifting fact keys, and two of your original roadmap items unbuilt.

34

02 Aug, 09:27
7 tools

A PDF of just our chats, for tracking.

Found Claude Code's own session log on disk and generated a 47-page transcript from the real record rather than from memory.

35

02 Aug, 09:31
4 tools

I wanted that summarised.

This document.

36

02 Aug, 09:50
0 tools

(switched model to Fable 5)

—

37

02 Aug, 09:50
0 tools

(command output)

—

Connecting her to the world

38

02 Aug, 09:52
34 tools

Three things: replies feel slow, she should say "you" rather than the name every time, and whether to add Groq for internet access given an unreliable connection.

Measured the real per-turn cost and found a genuine bug: the same sentence was being embedded **twice** per turn (memory and library lookups). Fixed by sharing one embedding call, and skipping retrieval entirely on short replies like "yeah". Softened the system prompt to prefer "you". Built weather (Open-Meteo, no key needed) and an online-lookup path via Groq — with the honest caveat that a plain Groq call has no more live internet access than the local model does either.

39

02 Aug, 09:52
0 tools

(switched model to Opus 5)

—

40**(command output)**

02 Aug, 09:52

—

0 tools

41**Handed over a Groq key, confirmed the media player is just "Windows," and asked about adding custom voice files.**

02 Aug, 10:20

33 tools

Set the key as a Windows environment variable, never in a file. While wiring it in, found the key had been pasted directly into `internet.py` and something had already stripped it back out, leaving broken syntax that would have crashed the whole app on next launch — fixed and verified. Built Windows media control (play/pause/skip/now-playing) via System Media Transport Controls, verified against a real session already running on the machine.

42**Asked whether voice model files could be used, and whether an OpenRouter key would help — then linked KittenTTS specifically for its "Bella" voice.**

02 Aug, 10:46

63 tools

Built OpenRouter support: with both keys present she now prefers OpenRouter's `:online` mode — a genuine web search — over Groq's plain guess. Getting Bella working meant fixing three separate upstream packaging bugs in KittenTTS (a required dependency version that doesn't exist on PyPI, a hardcoded Windows path to eSpeak, a data path baked in from the CI machine that built it). Verified working end to end, but left the active voice on amy since that was already the confirmed choice.

43**Provided the OpenRouter key, confirmed amy as the voice, and asked for a latency check plus updated PDFs.**

02 Aug, 11:30

37 tools

Verified OpenRouter's live search with a real, current, cited answer a local or Groq-only model couldn't have known. Measured all four internet paths cleanly: weather ~1.2s, connectivity check ~0.3s, Groq ~0.4–0.8s (fast, guesses), OpenRouter ~2.4–4.0s (slower, because it's actually searching). Confirmed the new trigger checks add no measurable cost to ordinary conversation. This document, and the three others.

44**Good work.**

02 Aug, 11:47

—

0 tools

What daily use surfaced

45**Asked when a restart is actually needed versus just running her file, and how long to wait before speaking after one.**

02 Aug, 11:52

29 tools

Found a real gap while answering: nothing stopped her running twice if launched by hand while the autostart copy was still up — two processes would fight over the same microphone. Added a Windows single-instance lock, tested across genuinely separate processes including surviving a forced kill, before answering the actual question: no restart needed, just run `Lia.exe`; wait for her audible greeting, that's the cue she's ready.

46**Provided two new keys after the leak, and asked to properly ensure the gitignore actually works.**

02 Aug, 12:25

8 tools

Set both new keys as environment variables and verified them working. Found the gitignore alone had done nothing, since `.claude/`, `build/` and `dist/` were already tracked in git — untracked 2019 files with `git rm --cached` without touching a single file on disk, so the gitignore could actually take effect from here on.

47

02 Aug, 13:50
42 tools

Five conclusions from actually living with her: CPU usage over 50%, music and alarms that didn't work, remove KittenTTS, rename her to Wade, and get web search genuinely working.

Measured the CPU complaint rather than guessing at it — the voice-detection library's ONNX runtime was spinning at ~290% of one core, continuously, while sitting completely idle. One environment variable fixed it, verified down to ~4% on the real packaged app. Built alarms and timers from nothing, including a background watchdog and persistence across restarts — caught and fixed a real parsing bug ("for 10 minutes" misread as a clock time) before it shipped. Made music commands admit honestly when there's nothing to play, after catching her invent a fake jazz playlist once during testing. Removed KittenTTS entirely, dropping the packaged app by 112MB. Renamed her to Wade. Broadened and verified the web-search path end to end with the new keys.

48

02 Aug, 14:17
0 tools

(a tool call was interrupted mid-turn)

—

49

02 Aug, 14:17
49 tools

Try again.

Re-ran the interrupted step and continued the same round of fixes.

50

02 Aug, 14:56
27 tools

Asked for the PDFs to be brought current, plus a new study reference covering the tools used in the project — NumPy and the rest.

This document, and the other three.

Learning to recognise her

51

02 Aug, 15:41
55 tools

Asked for a new "growth log" PDF tracking capability stages (making her, file-path reading, voices, and controlling actions as the current frontier) instead of a changelog — and, separately, whether she can actually control playback, since she'd told him her interactions were text-only.

Built the Growth Log and removed the old "What We Built" PDF as asked. On the capability question, found the real cause: her system prompt never told her what she could do, so she answered from generic assumptions instead of her actual abilities — confirmed directly in his own session log, where she'd said almost exactly that. Fixed the prompt, then built volume control as the natural next step. Tried simulating the physical volume keys first; verified live that it silently did nothing. Switched to talking to Windows' Core Audio API directly instead, verified working. Also caught and fixed a real matching bug along the way: "unmute" was resolving to the wrong command because character-level substring matching treated it as ending in "mute."

52

02 Aug, 16:14
3 tools

Asked to keep the PDFs updated going forward without being asked each time, and raised voice differentiation as a possible next step — asking for an opinion before any changes.

Saved the standing instruction. Recommended treating voice recognition as a soft comfort signal gating actions, not a hard lock — a laptop mic isn't reliable enough to bet security on, and a false lockout is worse than not having the feature at all.

53

02 Aug, 16:20
0 tools

Refined the idea: "hey Lia, it's Wade" as how she'd know him, and when she doesn't recognise the voice, she should be uneasy and decline actions.

Agreed on the design and flagged two decisions before building: what "uneasy" should actually sound like (a decline, not a hard refusal), and whether this should also gate shared personal facts, not just actions.

5402 Aug, 16:23
106 tools**Confirmed: gate personal details too, not just actions.**

Found a lightweight local model (3D-Speaker's CAM++ via sherpa-onnx, ~27MB, no PyTorch) and downloaded it. Building and testing it honestly took real debugging: an early result showed the enrolled voice scoring **worse** than a different one, traced through several false leads (averaging logic, database contamination) before finding the actual causes — Piper's TTS vocoder is stochastic by design, and a manual resampling step in the test harness was corrupting the signal. With those isolated, the model showed real, correctly-directioned discrimination, though weaker than expected — flagged clearly that synthetic test voices likely understate real accuracy, and it can't be fully verified without an actual human voice. Built enrollment ("it's Wade", averaged over repetitions), gated media/volume/alarm actions and personal facts behind a match, and added /whoami so the threshold can be tuned against real use rather than guessed at.

5502 Aug, 17:29
29 tools**Refined it further: a session-level passcode instead of a soft decline — "ENCODE" if she doesn't hear the identity phrase first, and stop listening entirely on a wrong answer. Separately, "DECODE" should have her explain the tools reference PDF.**

Rebuilt around exactly this. Flagged plainly that plain-text passcodes in `config.py` are visible to anyone if the repo is ever made public, the same caution as the earlier key leak. DECODE reads the PDF's full text directly rather than through the chunked library search, since "explain the whole thing" doesn't fit that retrieval pattern. Found and fixed two real bugs while testing live, not in theory: the new gate was intercepting "bye" before the exit check ever ran, which combined with a separate bug (the keyboard reader spinning with no backoff once stdin hit EOF) to produce a genuine infinite loop — over 600,000 identical lines in under a minute. Also caught that saying the identity phrase and a request in one breath ("it's Wade, play some music") silently lost the identity claim, because the command-rewriting step ran before the gate ever saw the original words.

5602 Aug, 17:52
58 tools**Try again.**

Re-ran the interrupted step and finished verifying both fixes live.

5702 Aug, 18:35
19 tools**Flag the passcode.**

Moved `PASSCODE_UNLOCK` to an environment variable (`LIA_PASSCODE`), the same pattern as the API keys, since a plaintext passcode in a possibly-public repo defeats its own purpose. Found a real edge case while fixing it: an *unset* passcode would have locked Wade out with something that could never be typed correctly — the gate now disables itself gracefully instead, with a one-time log line saying so. Verified both the configured and unconfigured cases live, and confirmed the fix didn't get recorded verbatim into git the way the original key leak did.

5802 Aug, 18:49
12 tools**Checked in that everything was still working, and asked for the PDFs and `requirements.txt` to be brought current.**

Audited `requirements.txt` against every actual import in the codebase (including the lazily-loaded ones inside functions, not just top-level) and found a real gap: `pypdf` — used by the library reader and now by DECODE too — had never been listed at all. A fresh install would have been missing it silently until the first PDF was opened. Added it. This document, and the rest.

What it came to

A companion called Wade knows by name, that starts at login, greets him, listens on an open mic, answers out loud, remembers across days, reads what he hands her, sets real alarms and timers, controls whatever's playing and the system volume, recognises his voice, and now gates every session behind either that or a spoken passcode — and never sends a word off the machine except the one narrow, optional exception she'll admit to: weather, and a question explicitly asked of the web. Thirteen modules, single-instance protected, running at roughly 4% of one core while idle instead of 290%.

Still open: no way to forget, she still occasionally invents small details, fact keys drift, there are no tests, and voice recognition's real-world accuracy is honestly unverified — tested only against synthetic voices, tunable via `/whoami` once there's real data to tune it against. Two real bugs (an access gate that could trap you in the conversation, an identity phrase silently lost when combined with a request) were found only by actually running the thing, not by reasoning about the code — the same lesson as the CPU issue and the alarm parsing bug before it.